MINI PROJECT – 20MCA245 | ACADEMIC PANEL PRESENTATION

AI-BASED ENTERPRISE DECISION SUPPORT PLATFORM USING MACHINE LEARNING

An Explainable AI-Driven Customer Churn Prediction & Sales Forecasting System

STUDENT DETAILS	PROJECT GUIDANCE
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INTRODUCTION: PROBLEM & MOTIVATION

1. Problem Domain Overview	2. Motivation Behind Selection
• **Subscription Economy:** Modern B2B SaaS, E-Commerce, and Finance rely heavily on recurring account revenue for growth. • **High Cost of Churn:** Acquiring a new customer costs **5x to 7x more** than retaining an existing account. A 5% churn reduction boosts profit by 25%–95%. • **Executive Decision Support:** Leadership requires automated platforms combining predictive scoring with actionable retention strategy. • **Proactive Risk Monitoring:** Legacy tools act after customer loss, whereas enterprise systems demand continuous risk identification.	• **Explainable AI (XAI):** Traditional ML models predict churn probability but fail to explain *why* specific accounts are leaving. • **Eliminating Black-Box Opacity:** Integrating SHAP TreeExplainer to calculate exact feature attributions for every prediction. • **Automated Prescriptive Engine:** Legacy dashboards alert managers of churn but offer no automated, personalized retention action playbooks. • **Unified Full-Stack Architecture:** End-to-end integration: Database → Random Forest / XGBoost → Risk Triage → Severity Root Causes → SHAP → React SPA & PDF Preview.

LITERATURE REVIEW: SELECTED RESEARCH PAPERS (PART 1)

Paper 1: Joy et al. (IEEE Access, 2024)	Paper 2: Bhuria & Aluvala (IEEE InCACCT, 2025)
• **Title:** A Big Data-Driven Hybrid Model for Enhancing Streaming Service Customer Retention Through Churn Prediction Integrated With Explainable AI • **Authors & Venue:** U. G. Joy, K. E. Hoque, M. N. Uddin, et al. (IEEE Access, 2024) • **Methodology:** Big Data hybrid ML architecture combined with SHAP feature attribution. • **Key Advantage:** Connects churn prediction directly with SHAP feature importance. • **Key Limitation:** Restricted to streaming domain; lacks automated prescriptive playbooks and live web application backend.	• **Title:** Predicting Bank Customer Churn: An XGBoost Approach to Enhancing Customer Retention • **Authors & Venue:** R. Bhuria, S. Aluvala (IEEE InCACCT, 2025) • **Methodology:** Multi-level ML framework evaluating KNN, XGBoost, Random Forest, and SVM. • **Key Advantage:** Empirical validation for XGBoost superiority in customer churn classification. • **Key Limitation:** Focuses strictly on accuracy; lacks SHAP explainability, root cause diagnosis, and action playbooks.

LITERATURE REVIEW: SELECTED RESEARCH PAPERS (PART 2)

Paper 3: Chandan et al. (IEEE ICRTCST, 2024)	Paper 4: Li & Yan (DSFE Journal, 2025)
• **Title:** A Novel Machine Learning-Based Early Warning Detection System for Business Customer Churn • **Authors & Venue:** R. R. Chandan, R. S., Sarishma, P. P. Devi, et al. (IEEE ICRTCST, 2024) • **Methodology:** Early warning detection architecture leveraging ML classifiers for risk identification. • **Key Advantage:** Establishes conceptual early-warning risk flow (Behavior → Risk → Action). • **Key Limitation:** Lacks post-hoc SHAP feature attribution, root cause diagnostic mapping, and unified database support.	• **Title:** Prediction of Bank Credit Customers Churn Based on Machine Learning and Interpretability Analysis • **Authors & Venue:** Y. Li, K. Yan (Data Science in Finance and Economics, 2025) • **Methodology:** XGBoost classifier + SHAP interpretability for bank credit churn detection. • **Key Advantage:** Evaluates individual variable influences using SHAP (~97% AUC). • **Key Limitation:** Purely diagnostic research study; lacks automated prescriptive action playbooks and enterprise web platform.

LITERATURE REVIEW: COMPARATIVE ANALYSIS MATRIX

Paper / Reference System	ML Model	Risk Triage	SHAP / XAI	Severity Root Cause	Prescriptive Playbooks	Unified Full-Stack SPA & PDF Preview
Paper 1: Joy et al. (IEEE 2024)	ML Hybrid	✓ Yes	✓ SHAP	✕ Partial	✕ No	✕ No
Paper 2: Bhuria & Aluvala (IEEE 2025)	XGBoost / RF	✓ Yes	✕ No	✕ No	✕ No	✕ No
Paper 3: Chandan et al. (IEEE 2024)	ML Classifier	✓ Early Warning	✕ No	✕ No	✕ No	✕ No
Paper 4: Li & Yan (DSFE 2025)	XGBoost	✓ Yes	✓ SHAP	✓ Yes	✕ No	✕ No
PROPOSED SYSTEM (Joyal Jobai)	**RF & XGBoost**	**✓ High/Med/Low**	**✓ SHAP Engine**	**✓ Severity-Ranked**	**✓ Automated Playbooks**	**✓ FastAPI + React SPA & PDF Preview**

LITERATURE REVIEW: IDENTIFIED RESEARCH GAP & NOVELTY

1. Research Gap & Theoretical Progression	2. The Core Innovation & Novelty
• **Fragmented State-of-the-Art:** Prior research treats churn scoring, SHAP feature attributions, and early warning systems as isolated theoretical techniques. • **Diagnostic Void:** Existing academic frameworks stop at raw prediction probabilities or static plots, failing to translate model output into automated retention playbooks. • **Static Output Limitations:** Literature lacks customer-specific primary driver ranking, returning identical generic statements for high-risk accounts.	• **End-to-End Platform Integration:** Seamlessly connects SQLite/MySQL Database → Random Forest / XGBoost ML Inference → High/Med/Low Risk Triage → Severity-Ranked Root Cause Extractor → SHAP Explainability → Prescriptive Retention Action Engine → React SPA & Live PDF Preview. • **Dynamic Severity Ranking:** Ranks customer friction vectors mathematically to surface personalized primary drivers. • **Interactive PDF Preview Integration:** Enables instant browser preview (`window.open(url, '_blank')`) without force-downloading files.

EXISTING SYSTEM: OPERATIONAL OVERVIEW & LIMITATIONS

1. Existing Approaches in Enterprise	2. Critical Operational Limitations
• **Traditional BI Dashboards:** Static reporting tools (Tableau/PowerBI) display historical churn numbers after accounts have already cancelled. • **Manual Spreadsheet Audits:** Account managers review customer logs manually using basic Excel filtering and subjective guesswork. • **Isolated Statistical Models:** Standalone logistic regression scripts operate independently of production CRM databases. • **Unstructured Retention Process:** Customer intervention relies on ad-hoc phone calls without standardized risk playbooks.	• **Reactive Detection:** Churn is detected after accounts have ceased purchasing, missing early intervention windows. • **Black-Box AI Opacity:** ML models output raw risk scores (e.g., 85% risk) without explaining individual feature drivers. • **Manual Investigation Delays:** Teams must manually investigate churn drivers, delaying outreach by days or weeks. • **Data & Workflow Fragmentation:** Databases, predictive models, and retention steps exist in silos, incurring 5x–7x re-acquisition costs.

PROPOSED SYSTEM: OVERVIEW & KEY PLATFORM MODULES

High-Level Platform Architecture & Modules

1. **Database Ingestion Layer:** SQLite / MySQL relational storage for customer profiles, transaction history, and prediction logs.
2. **ML Predictive Engine:** Scikit-Learn Random Forest Classifier (v1.0) & XGBoost scoring risk with **94.87% accuracy**.
3. **Risk Triage Classification:** Automatically categorizes accounts into High Risk (≥70%), Medium Risk (30-69%), and Low Risk (<30%).
4. **Dynamic Severity Root Cause Engine:** Evaluates customer friction signals (support tickets, late deliveries, inactivity days, cart abandonments, ratings) and ranks them mathematically.
5. **SHAP TreeExplainer (XAI):** Computes exact local Shapley values showing exact positive and negative feature attributions.
6. **Prescriptive Retention Playbooks:** Automatically maps friction vectors to targeted action plans (priority support, win-back discounts).
7. **React SPA & Live PDF Preview:** Interactive executive dashboard featuring single page navigation and browser PDF preview (`window.open(url, '_blank')`).

SYSTEM ARCHITECTURE: DATA FLOW PIPELINE

1. Data Ingestion (SQLite / MySQL)	2. ML Inference (Random Forest)	3. Risk Triage (High/Med/Low)	4. Severity Root Cause (Diagnostic Engine)	5. SHAP XAI (TreeExplainer)	6. Prescriptive Action (Decision Engine)	7. Dashboard & PDF (React + ReportLab)

Pipeline Stage Execution Breakdown

• **Data Preprocessing & Scaling:** Customer records are ingested, missing values imputed via SimpleImputer, and continuous variables normalized via StandardScaler.
• **Inference & Risk Triage:** Random Forest Classifier predicts churn probability ($P(\text{Churn})$) and categorizes accounts into High, Medium, or Low Risk.
• **Severity-Ranked Root Cause Analysis:** Friction metrics are evaluated against mathematical severity formulas to identify the #1 primary cause per customer.
• **SHAP Feature Attribution:** TreeExplainer calculates exact local Shapley values (\phi_i) quantifying exact feature contributions.
• **Prescriptive Action & Presentation:** Action playbooks are generated and exposed via FastAPI REST endpoints to the React SPA dashboard and ReportLab PDF preview generator.

METHODOLOGY & CORE ML/XAI MODULES

Predictive Analytics & XAI Modules	Diagnostic & Prescriptive Engine Modules
• **Random Forest Classifier (v1.0):** Ensemble of 100 decision trees (max depth = 12) trained with balanced class weights for robust churn probability scoring. • **XGBoost Classifier:** Baseline gradient boosted trees model optimizing binary cross-entropy loss. • **SHAP TreeExplainer:** Computes local Shapley values breaking down feature contributions (+/-) for every customer. • **Scikit-Learn Pipeline:** Clean ColumnTransformer handling categorical OneHotEncoding and numerical scaling.	• **Dynamic Severity Root Cause Extractor:** Ranks friction metrics (support tickets, late deliveries, inactivity days, cart abandonments, ratings) mathematically to derive personalized primary drivers. • **Prescriptive Action Engine:** Rule-driven recommendation algorithm mapping primary causes to retention playbooks. • **Executive React SPA & ReportLab PDF Engine:** Modern single-page web dashboard and server-side PDF generator with live browser preview.

TECHNOLOGY STACK & DEVELOPMENT ENVIRONMENT

Category	Technologies Used	Project Application & Purpose
Programming	Python 3.12, JavaScript (ES6+), SQL, HTML5/CSS3	Backend API logic, ML pipelines, SHAP calculations, and React frontend dashboard UI.
ML & Data Science	Random Forest, XGBoost, SHAP, Scikit-Learn, Pandas, NumPy	Classification modeling, feature transformation, SHAP TreeExplainer, data manipulation.
Backend & ORM	FastAPI, Uvicorn, SQLAlchemy ORM, Pydantic v2	High-performance asynchronous REST API server, CORS handling, data schema validation.
Frontend UI	React 18, Vite, Recharts, Lucide React Icons, Axios	Single Page Application (SPA), executive KPI cards, customer inspection modals, PDF preview.
Database & Reports	SQLite, MySQL 8.0, PyMySQL, ReportLab Engine	Relational storage for customer profiles, orders, churn predictions, and dynamic PDF reports.

SPRINT PLANNER: AGILE DEVELOPMENT ROADMAP

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SPRINT 1 (WEEKS 1-3) — COMPLETED ✓	SPRINT 2 (WEEKS 4-6) — COMPLETED ✓
• **Major Tasks:** Requirements gathering, dataset cleaning, Random Forest & XGBoost model training, SQLAlchemy schema initialization. • **Deliverables:** Model artifacts (`random_forest_model.joblib`, `preprocessor.joblib`), connected database schemas. • **Status:** 100% Completed & Verified.	• **Major Tasks:** SHAP TreeExplainer integration, dynamic severity root cause engine, ReportLab PDF generation service, FastAPI REST routing. • **Deliverables:** Functional XAI API backend, automated severity-ranked primary drivers, ReportLab PDF engine. • **Status:** 100% Completed & Verified.
SPRINT 3 (WEEKS 7-9) — COMPLETED ✓	**SPRINT 4 (WEEKS 10-12) — ACTIVE ■**
• **Major Tasks:** React + Vite Single Page Application, Executive Dashboard, Customer Portfolio Directory, Account Inspection Modal, PDF preview integration. • **Deliverables:** Production-ready web dashboard with live `window.open()` PDF preview. • **Status:** 100% Completed & Deployed.	• **Major Tasks:** End-to-end integration testing, query optimization, governance documentation, final presentation panel preparation. • **Deliverables:** Integrated Decision Support Platform & final presentation documentation. • **Status:** Active Final Phase.

SPRINT RESULTS & MODEL PERFORMANCE METRICS

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94.87%	0.954	0.910	0.890
CLASSIFICATION ACCURACY	AUC-ROC SCORE	PRECISION SCORE	RECALL SCORE

Sprint Deliverables Achieved	Current Project Operational Status
✔ **10,000+ Records Preprocessed:** Handled missing value imputation, categorical one-hot encoding, and standard scaling. ✔ **Random Forest & XGBoost Deployed:** 100% operational inference pipeline trained on customer telemetry. ✔ **Dynamic Severity Root Causes:** Eliminates static text duplication by mathematically ranking top friction vectors. ✔ **Live PDF Preview Integration:** Enabled instant browser PDF preview (`window.open(url, '_blank')`) across all dashboard views.	• **FastAPI REST Backend:** 100% operational REST endpoints for authentication, customer directory, model info, and PDF reports. • **React SPA Frontend:** Polished Single Page Application featuring interactive KPI cards, account profile inspection modals, and live PDF preview. • **Validation Check:** All model metrics and PDF outputs verified with zero boundary overflow issues.

EXPECTED OUTCOMES & FUTURE SCOPE

Expected Platform Impact	Real-World Applications	Scope for Future Scope
• **15–20% Churn Reduction:** Proactive early warning risk triage enables timely account retention. • **<2 sec Explainability:** Instant SHAP feature attributions and automated severity root cause analysis. • **Actionable Playbooks:** Replaces guesswork with automated retention recommendations.	• **B2B SaaS Enterprises:** Subscription retention management and automated customer success workflows. • **E-Commerce Platforms:** Customer drop-off detection, return friction reduction, and win-back offers. • **Banking & Telecom:** Account churn monitoring, deposit retention, and proactive contract renewal.	• **Real-Time Streaming:** Apache Kafka streaming pipeline for instant transaction telemetry ingestion. • **LLM Decision Summaries:** Generative AI natural language executive briefing reports. • **Multi-Tenant Cloud:** Dockerized microservices architecture with AWS/GCP deployment.

REFERENCES & ACADEMIC BIBLIOGRAPHY

Academic Literature Citations (IEEE Format)

[1] U. G. Joy, K. E. Hoque, M. N. Uddin, L. Chowdhury, and S.-B. Park, "A Big Data-Driven Hybrid Model for Enhancing Streaming Service Customer Retention Through Churn Prediction Integrated With Explainable AI," *IEEE Access*, vol. 12, pp. 69130–69150, 2024, doi: 10.1109/ACCESS.2024.3401247.

[2] R. Bhuria and S. Aluvala, "Predicting Bank Customer Churn: An XGBoost Approach to Enhancing Customer Retention," in *Proc. 2025 3rd International Conference on Advancement in Computation & Computer Technologies (InCACCT)*, IEEE, 2025, doi: 10.1109/InCACCT65424.2025.11011351.

[3] R. R. Chandan, S. Ramaraj, Sarishma, and P. P. Devi, et al., "A Novel Machine Learning-Based Early Warning Detection System for Business Customer Churn," in *Proc. 5th International Conference on Recent Trends in Computer Science and Technology (ICRTCST)*, IEEE, 2024, doi: 10.1109/ICRTCST61793.2024.10578489.

[4] Y. Li and K. Yan, "Prediction of Bank Credit Customers Churn Based on Machine Learning and Interpretability Analysis," *Data Science in Finance and Economics*, vol. 5, no. 1, pp. 25–42, 2025, doi: 10.3934/DSFE.2025002.